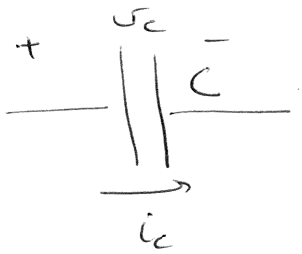
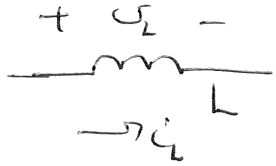
Chapter 9. Lecture 1.

→ Revision:





* $i_c = C \dfrac{dv_c}{dt}$
* $v_L = L \dfrac{di_L}{dt}$
* $C = q/v$
* $w_c = \frac{1}{2} C v_c^2$
* $w_L = \frac{1}{2} L i_L^2$.
* DC → open circuit
* DC → closed circuit.

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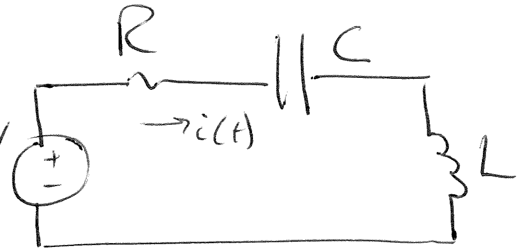
* Consider the following circuit:
  → Solve for $i(t)$.

$v_s = 10\cos 4t$ V



* KVL: $-v_s + i(t)R + v_c + v_L = 0$

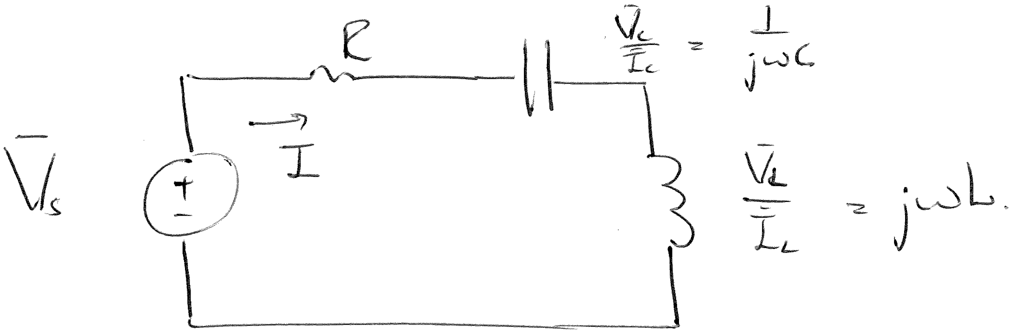
$$-v_s + i(t)R + \frac{1}{C}\int_{t_0}^{t} i(t)\,dt + L\frac{di(t)}{dt} = 0$$

$\to \dfrac{d}{dt}$:

$$\frac{d}{dt} i(t)R + \frac{1}{C} i(t) + L\frac{d^2 i(t)}{dt^2} = \frac{d}{dt} v_s.$$

* What now?

* Consider the same circuit where all reference to time has been removed.



Time	$\longleftrightarrow$	Phasor
$v_s = 10\cos(4t)$		$\bar{V}_s = 10\angle 0^\circ$
$i(t)$		$\bar{I}$
R		R
$i_c = C\frac{dv_c}{dt}$		$\bar{I}_c = j\omega C\,\bar{V}_c$
$v_L = L\frac{di}{dt}$		$\bar{V}_L = j\omega L\,\bar{I}_L$

$\therefore$ KVL: $-\bar{V}_s + \bar{I}R + \bar{V}_c + \bar{V}_L = 0$

$$\bar{I}R + \frac{\bar{I}}{j\omega C} + j\omega L\bar{I} = \bar{V}_s$$

$$\therefore \bar{I} = \frac{\bar{V}_s}{R + j\omega L + \frac{1}{j\omega C}}$$

* $\bar{I}$ has been calculated in the phasor domain.
* It can be transformed back to find the time domain solution.

* Sinusoids:

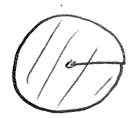
→ Sinusoid: cos(θ) or sin(θ) waveform.

→ General expression:

$$v(t) = V_m \cos(\omega t + \phi)$$

V_m	→	amplitude.	[V] (or [A])
ω	→	angular frequency	[rad/s]
t	→	time	[s]
ϕ	→	phase angle	[rad].

* Radians vs degrees:

→ Circle circumference = $2\pi r$  360°

→ $\frac{1}{2}$ Circle circum = πr  180°

→ $\frac{1}{5}$ Circle circum = $\frac{2}{5}\pi r$  72°

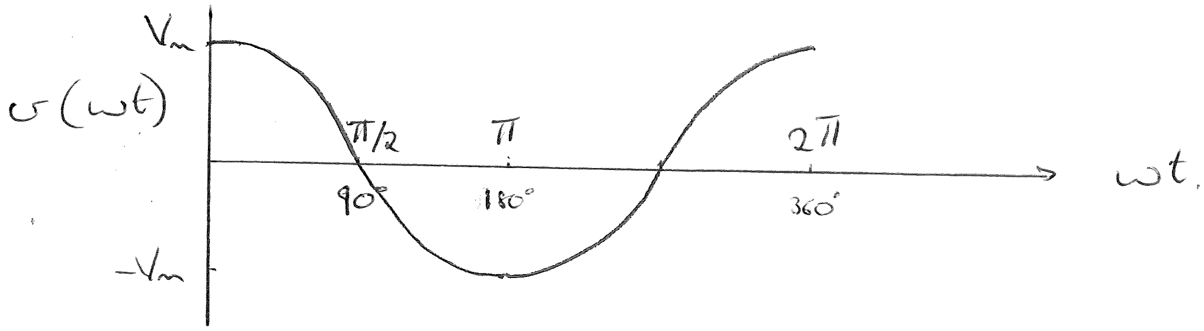
∴ arc length = radians × radius.

* Relationship between ° and rad.

$$\boxed{180° = \pi \text{ rad.}}$$

* <u>Sinusoids:</u> <u>ω and t</u>

$$\rightarrow v(t) = V_m \cos(\omega t).$$



$$v(0) = V_m = V_m \cos(0)$$

$$v\left(\omega t = \frac{\pi}{2}\right) = 0 = V_m \cos\left(\frac{\pi}{2}\right)$$

$$v(\omega t = 2\pi) = V_m = V_m \cos(2\pi)$$

* But, if $\omega t = 2\pi$, then a full period T. of the wave has passed.

$\therefore \quad t = T$ when $\omega t = 2\pi$

$\therefore \quad \omega T = 2\pi \qquad (t = T)$

$$\omega = \frac{2\pi}{T}$$

but $f = \frac{1}{T}$ $\quad \boxed{\therefore \ \omega = 2\pi f.}$

$f \rightarrow$ frequency : rotations per second. [Hz]

$2\pi \rightarrow$ radians per rotation [rad]

$T \rightarrow$ seconds per rotation [s]

$\omega \rightarrow$ angular frequency [rad/s].

* <u>Sinusoids</u> <u>$\phi \rightarrow$ the phase angle.</u>

* $v(t) = V_m \cos(\omega t + \phi)$

→ If $\phi > 0$, the graph moves left

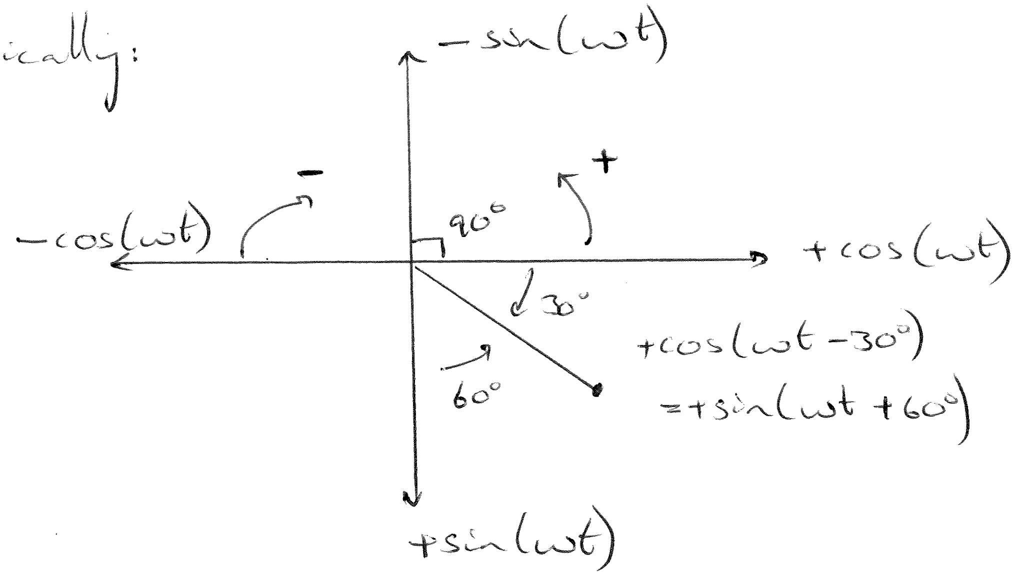
→ If $\phi < 0$, the graph moves right.

$\therefore$ ϕ changes the <u>phase</u> of the sinusoid.

* <u>Important identities:</u>

$$\sin(\omega t \pm 180^\circ) = -\sin(\omega t)$$
$$\cos(\omega t \pm 180^\circ) = -\cos(\omega t)$$
$$\sin(\omega t \pm 90^\circ) = \pm \cos(\omega t)$$
$$\cos(\omega t \pm 90^\circ) = \mp \sin(\omega t)$$

The identities above can be illustrated graphically:



Examples

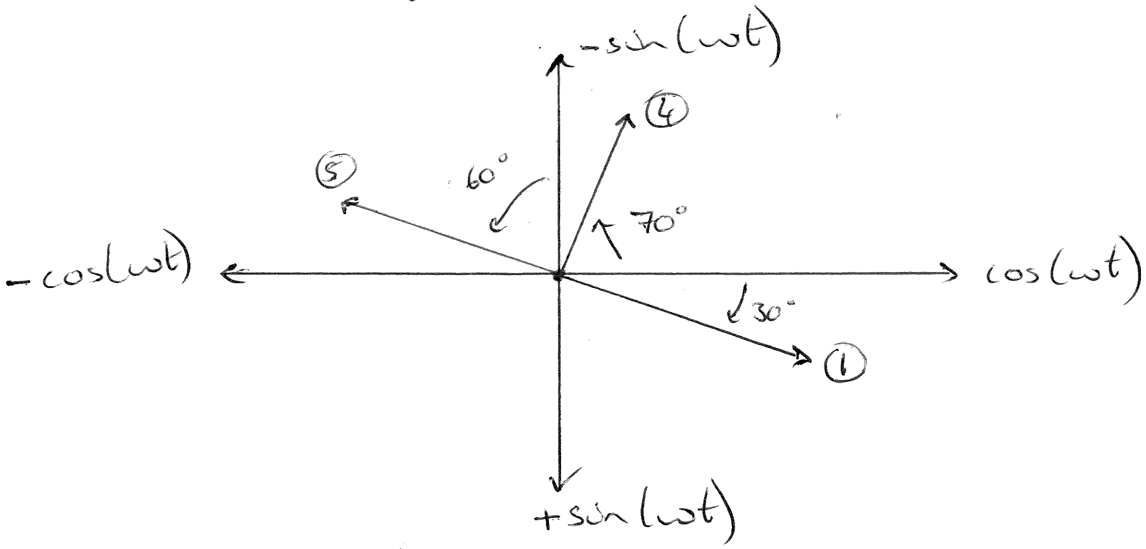
① $\cos(\omega t - 30°) = \sin(\omega t \ldots$

② $-\cos(\omega t + 110°) = \sin(\omega t \ldots$

③ $\sin(\omega t - 60°) = \cos(\omega t \ldots$

④ $\sin(\omega t + 160°) = \cos(\omega t \ldots$

⑤ $-\sin(\omega t + 60°) = \cos(\omega t \ldots$



→ Determine the phase angle between:

$$i_1 = -4\sin(377t + 25°)$$

$$i_2 = 5\cos(377t - 40°)$$

$$\boxed{\text{Always give angle: } 0 \leq \phi \leq 180°}$$

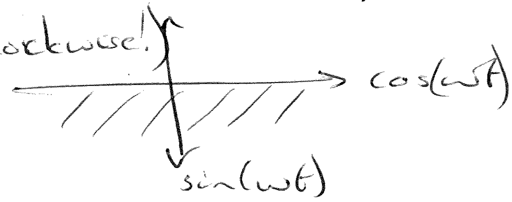
* Leading and lagging.

→ Lead → signal "a" reaches maximum before signal "b" wrt time.

→ Lag → signal "a" reaches maximum after signal "b" wrt time.

* Rules

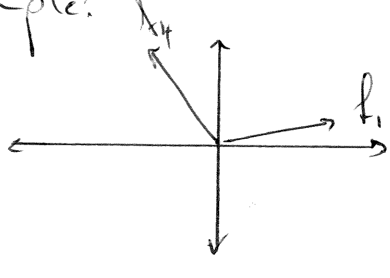
① Rotate both vectors so that both are in the bottom two quadrants.
(Always anti-clockwise!)



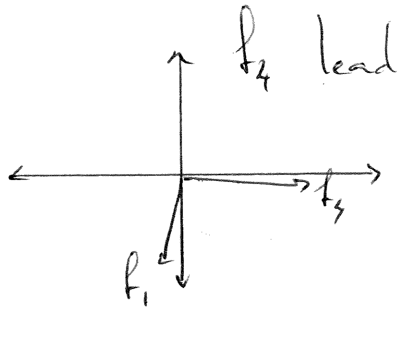
② Once both vectors are in bottom quadrants, continue to rotate ~~until~~ anti-clockwise until one cuts + cos(wt) axis.

③ This vector leads the other by an angle between 0° and 180°.

Example:



⇒



f_4 leads f_1

Examples.

$$f_1 = \cos(\omega t + 0^\circ)$$

$$f_2 = \cos(\omega t + 110^\circ)$$

$$f_3 = \cos(\omega t - 40^\circ)$$

$$f_4 = \cos(\omega t - 110^\circ)$$

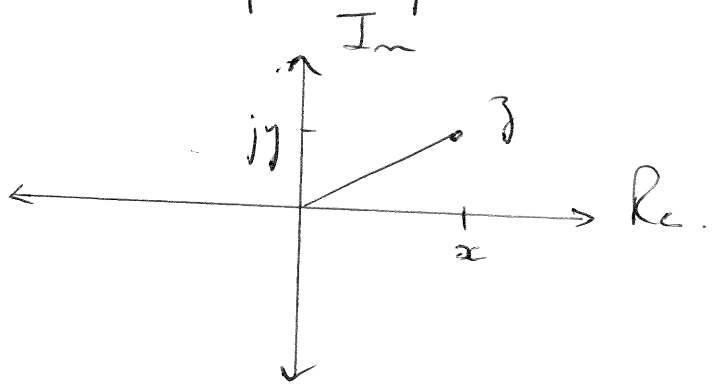
By how many degrees is f_2 leading or lagging the other sinusoids.

Chapter 9: Lecture 2

* Complex numbers

$$\rightarrow \underset{\substack{\downarrow \\ \mathbb{C} \\ \text{(Complex)}}}{z} = \underset{\substack{\downarrow \\ Re \\ \text{(Real)}}}{x} + \underset{\substack{\downarrow \\ Im \\ \text{(Imaginary)}}}{jy} \qquad \boxed{j = \sqrt{-1}}$$

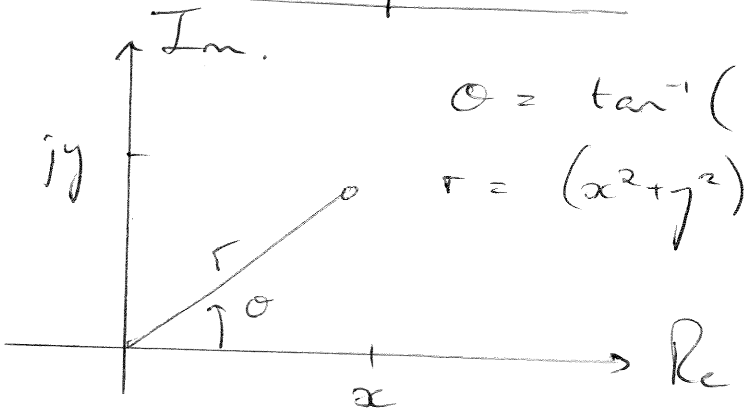
* Every complex number has a real and imaginary part. This can be shown on the complex plane:



* Complex numbers can be written in 3 different ways:

1. Rectangular : $z = x + jy$
2. Polar : $z = r \angle \theta$
3. Exponential : $z = re^{j\theta}$

* Rectangular, Polar, Exponential.



$$\theta = \tan^{-1}(y/x)$$
$$r = (x^2+y^2)^{1/2}.$$

① (Rectangular) $z = x + jy$

$$= r(\cos\theta + j\sin\theta)$$

② (Polar) $z = r\angle\theta$ where $r = \sqrt{x^2+y^2}$ and $\theta = \tan^{-1}(y/x)$

③ (Exponential) $z = re^{j\theta}$ where:

Euler's formula:

$$e^{j\theta} = \cos\theta + j\sin\theta \quad (= 1\angle\theta).$$

Remember:

$$1\angle\phi = e^{j\phi} = \cos\phi + j\sin\phi$$
$$1\angle-\phi = e^{-j\phi} = \cos(-\phi) + j\sin(-\phi)$$
$$= \cos(\phi) - j\sin(\phi)$$

$$\therefore \quad 1\angle\pm\phi = e^{\pm j\phi} = \cos\phi \pm j\sin\phi$$

* Arithmetic and complex numbers.

① Addition: (Rectangular)

$$z_1 + z_2 = (x_1 + jy_1) + (x_2 + jy_2)$$
$$= (x_1 + x_2) + j(y_1 + y_2)$$

② Subtraction: (Rectangular)

$$z_1 - z_2 = (x_1 + jy_1) - (x_2 + jy_2)$$
$$= (x_1 - x_2) + j(y_1 - y_2)$$

③ Multiplication (Rectangular or polar)

$$z_1 \times z_2 = (x_1 + jy_1)(x_2 + jy_2)$$
$$= x_1 x_2 + jx_2 y_1 + jx_1 y_2 + j^2 y_1 y_2$$
$$= x_1 x_2 + jy_1 x_2 + jx_1 y_2 - y_1 y_2$$
$$= (r_1 \angle \theta_1)(r_2 \angle \theta_2) = r_1 e^{j\theta_1} r_2 e^{j\theta_2}$$
$$= r_1 r_2 \angle(\theta_1 + \theta_2) = r_1 r_2 e^{j(\theta_1 + \theta_2)}$$

④ <u>Division:</u> (Polar)

$$\frac{r_1 \angle \theta_1}{r_2 \angle \theta_2} = \frac{r_1 e^{j\theta_1}}{r_2 e^{j\theta_2}}$$

$$= \frac{r_1}{r_2} e^{j(\theta_1 - \theta_2)}$$

$$= \frac{r_1}{r_2} \angle \theta_1 - \theta_2$$

⑤ <u>Root:</u> (Polar / Exponential)

$$\sqrt{z} = \sqrt{r \angle \phi}$$

$$= \sqrt{r e^{j\phi}}$$

$$= \sqrt{r} \left(e^{j\phi}\right)^{1/2}.$$

$$= \sqrt{r} \angle \phi/2.$$

⑥ Inverse: (Polar / Exponential)

$$1/z = \frac{1}{r \angle \phi} = \frac{1}{r e^{j\phi}}$$

$$= \frac{1}{r} e^{-j\phi} = \frac{1}{r} \angle -\phi$$

⑦ Conjugate: $z^* = (x + jy)^* = x - jy$

$$= (r \angle \phi)^* = r \angle -\phi$$

Examples

(1) Draw these numbers on the complex plane and convert to polar format

$j = \ldots \angle \ldots$

$-j = \ldots \angle \ldots$

$-1 =$

$+1 =$

$1+j =$

$1-j =$

$-1+j =$

$-1-j =$

(2) Calculate:

$$\left[(5+j2)(-1+j4) - 5\angle 60^\circ\right]^*$$

(3) Calculate:

$$\frac{10+j5+3\angle 40^\circ}{-3+j4} + 10\angle 30^\circ + j5$$

Chapter 9 Lecture 3:

Aim: $A\cos(\omega t + \phi) \Leftrightarrow A\angle\phi$

Time domain Complex domain.

* Consider the time signal $v(t)$ and Euler's formula:

$$\rightarrow v(t) = V_m \cos(\omega t + \theta)$$

$$\rightarrow e^{j\phi} = \cos\phi + j\sin\phi$$

$$\therefore e^{j(\omega t + \theta)} = \cos(\omega t + \theta) + j\sin(\omega t + \theta)$$

→ We can write $v(t)$ as follow:

$$v(t) = Re\left[e^{j(\omega t + \theta)} \times V_m\right]$$

$$= Re\left[V_m e^{j\omega t} e^{j\theta}\right]$$

$$= Re\left[\underbrace{V_m\angle\theta}_{\text{Phasor}} \times \underbrace{1\angle\omega t}_{\text{Time component.}}\right]$$

→ What is a **phasor**?

* A phasor is a rotating vector.
* It has both magnitude and direction.
  (amplitude) (phase)

→ How does a **phasor** rotate?

* Rotation comes from "ωt".
* All information contained in V_m and ϕ.

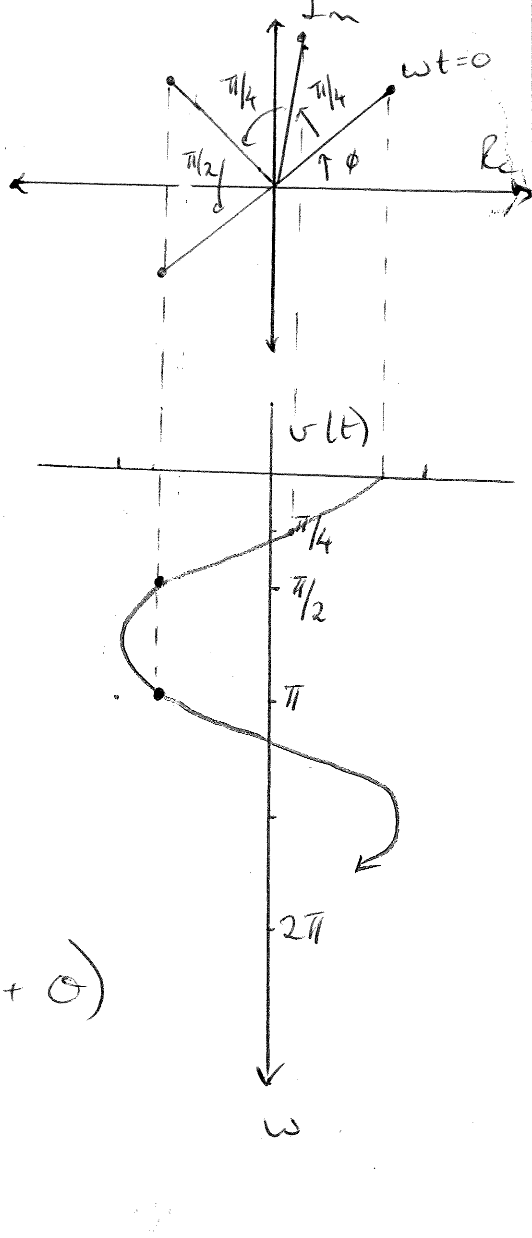
$$v(t) = V_m \cos(\omega t + \theta)$$
$$= \mathrm{Re}\left[V_m \angle\theta \times 1\angle\omega t\right]$$

→ $\omega t = 0$: $v(t) = \mathrm{Re}\left[V_m \angle \theta + 0^\circ\right]$

→ $\omega t = \pi/4$: $v(t) = \mathrm{Re}\left[V_m \angle \theta + \pi/4\right]$

→ $\omega t = \pi/2$: $v(t) = \mathrm{Re}\left[V_m \angle \theta + \pi/2\right]$

→ $\omega t = \pi$: $v(t) = \mathrm{Re}\left[V_m \angle \theta + \pi\right]$



$\therefore$ ωt → rotation at rad/s

$\left.\begin{array}{l} V_m \rightarrow \text{amplitude} \\ \theta \rightarrow \text{phase} \end{array}\right] \rightarrow$ Remains constant.

$\therefore$ Time: $v(t) = V_m \cos(\omega t + \theta)$

Phasor $\bar{V} = V_m \angle\theta$